

Connor Parrow

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PROFESSIONAL SUMMARY

- Fifth-year Ph.D. candidate in Applied and Computational Mathematics.
- Specialized in high-order accuracy simulation numerical methods of fluid–structure interaction and multiphysics systems.
- Experienced with finite element methods, thermal-fluids modeling, and stability/error analysis of simulation algorithms.
- Strong background in interdisciplinary research bridging applied mathematics, mechanical/biological systems, and computational engineering.

EDUCATION

University of Notre Dame

South Bend, IN

M.S., Applied and Computational Mathematics,

January 2025

Ph.D, Applied and Computational Mathematics,

Expected May 2027

Hobart and William Smith Colleges

Geneva, NY

B.S., Mathematics and Biology; GPA: 3.96, *summa cum laude*

May 2021

PUBLICATIONS

- Parrow, C., Mangini, A., Bukač, M. "A partitioned, second-order method for two-phase flow." *Submitted to Advances in Computational Mathematics* (July 2026).
- Parrow, C., Bukač, M. "Stability and Error Analysis of a Partitioned Robin-Robin Method for Fluid Poroelastic Structure Interactions." *Journal of Scientific Computing* 106.2 (2026):40. <https://doi.org/10.1007/s10915-025-03143-1>
- Parrow, Connor, and Martina Bukač. "A Robin–Robin strongly coupled partitioned method for fluid–poroelastic structure interaction." *Journal of Numerical Mathematics* 33.3 (2025): 289-312. <https://doi.org/10.1515/jnma-2024-0057>
- Bell, J.R., Nwobbi, L., Parrow, C., Wheeler, A., & Zelleke, E. "Graceful wind chimes." *Involve, a Journal of Mathematics* 14.4 (2021): 683-696.

PROFESSIONAL EXPERIENCE

University of Notre Dame

South Bend, IN

Research/Teaching Assistant, Applied and Computational Mathematics and Statistics

August 2022 – Present

- Developed and analyzed Finite Element models for Fluid-Poroelastic Structure Interaction problems, establishing algorithmic stability and second-order convergence.
- Designed and executed computational experiments analogous to structural/thermal testing, validating new simulation approaches against theory resulting in 3 journal-level research papers.
- Taught programming labs in Python and RStudio, mentoring students in numerical modeling and data analysis.

Instructor of Record, ACMS 20340 - Statistics for Life Sciences

Spring 2026

- Helped students learn to critically evaluate information and apply logical analysis in order to make proper decisions on statistical methods to use in life and health sciences problems.
- Taught the importance of analytical models such as discrete and continuous probability distributions, sampling distributions, confidence intervals, hypothesis testing and other statistical methods.

Undergraduate Research Project Mentor

January 2025-May 2026

- Guided two undergraduate students in developing simulation algorithms for the Cahn-Hilliard Navier-Stokes equations to study multiphase fluid dynamics in 5 numerical experiments.

enFocus

South Bend, IN

Civil Innovation Intern

June 2026 – July 2026

- Designed and implemented sampling methodology and call trees for an access audit as part of a joint Community Health Needs Assessment, collecting data on patient experiences accessing healthcare by phone.

EDGE Foundation

Knoxville, TN

Program Mentor

June 2024 – July 2024

- Organized problem sessions for and mentored 20 incoming Ph.D. students in mathematics across the country, covering topics such as Abstract Algebra, Real Analysis, and Measure Theory.

AWARDS AND HONORS

- Advance Research Award Honorable Mention Award April 2026
- Graduate School Professional Development Award March 2025
- Graduate Student Government Professional Development Award April 2025
- ACMS Department Travel Award June 2025, Jan. 2024
- SIAM AN24 Travel Award July 2024
- SIAM EDGE Participant Fellow July 2022
- Gilman Scholar

PRESENTATIONS

- Conference Presentation, *Midwest Numerical Analysis Day* April 2026
- Conference Presentation, COSE Research Horizons Symposium December 2025
- Invited Discussion Leader, *Mathematics behind Vaccine Policies-Science Policy Journal Club* November 2025
- Invited Speaker, *Association For Women in Science STEM Luncheon* November 2025
- Invited Speaker, *International Conference on Spectral and High Order Methods* July 2025
- Invited Speaker, *30th Biennial Numerical Analysis Conference* June 2025
- Conference Presentation, *Joint Math Meetings, Research by Women of EDGE Special Session* January 2025
- Poster, *Graduate Research Symposium* April 2025
- Poster, *AWM Special Session at SIAM AN24* July 2024
- Poster, *Mathematical Models and Numerical Methods for Multiphysics Systems Conference* April 2024

LEADERSHIP & SERVICE

Association for Women in Mathematics (AWM), Student Chapter President August 2024 – Present

- Plan workshops and outreach programs promoting women in mathematics including over 20 members.

Graduate Student/PostDoc Seminar, Cofounder and Co-organizer February 2025 – Present

- Organize biweekly seminars for graduate students and postdocs to prepare for giving talks at conferences.

Society for Industrial and Applied Mathematics (SIAM), Chapter President May 2025 – Present

Society for Industrial and Applied Mathematics (SIAM), Chapter Treasurer August 2024 – May 2025

- Work with board members to plan events involving community outreach and career development.

Graduate Student Government, ACMS Department Representative August 2023 – May 2024

- Represented student interests on the Quality of Life subcommittee and advocated for coverage of dependents on graduate students health insurance.

TECHNICAL & SOFT SKILLS

- Modeling & Simulation:** Finite Element Methods, Multiphysics Simulation, CFD, Multiphase flow
- Programming & Tools:** MATLAB, Python, C++, FreeFEM++, RStudio, LaTeX
- Domain Expertise:** Fluid Dynamics, Fluid–Structure Interaction, Numerical Analysis, PDEs, Machine Learning
- Soft Skills:** Mentoring & Teaching, Collaborative Research, Scientific Writing, Adaptability